

Kareem Muftee

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EDUCATION

University of Illinois at Chicago
Bachelor of Science in Computer Science
Coursework: Data Structures, Systems Programming, Software Engineering I, Computer Architecture, Databases, Software Design

Expected May 2027

TECHNICAL SKILLS

Programming Languages: C++, Go, TypeScript, JavaScript, Python, SQL
Backend/Systems: Node.js, Express, FastAPI, REST APIs
Data/Streaming: Redis, Kafka, Redpanda, PostgreSQL, OpenSearch
Infrastructure: Docker, NGINX, Linux, AWS (S3, EC2), OVHcloud

PROJECTS

Real-Time News Aggregation System
• Engineered a low-latency news aggregation platform processing live RSS feeds through an event-driven pipeline using WebSockets, Kafka/Redpanda, and Redis caching.
• Designed a scalable backend architecture with decoupled ingestion and indexing workers, enabling fault-tolerant data streaming and real-time updates.
• Integrated OpenSearch for full-text indexing and high-speed querying across large datasets, improving retrieval performance and filtering capabilities.
• Deployed services on a self-hosted VPS using Docker and NGINX, simulating production-like infrastructure and system design patterns.

Meteor Simulation System
• Built a high-performance simulation system leveraging parallel computation to process large-scale scientific datasets efficiently.
• Integrated multiple real-world datasets (NASA NEO, USGS, GEBCO) to model realistic environmental and seismic effects.
• Developed an asynchronous backend using FastAPI and exposed APIs for real-time interaction and control.

Structured Text Rendering Engine
• Designed a custom rendering system mapping structured text data to precise layouts using a SQLite-backed data pipeline.
• Implemented efficient word-level indexing and interaction logic, enabling dynamic selection without breaking layout constraints.
• Optimized rendering consistency across edge cases, improving performance and alignment accuracy.

June 2025 - Present

Oct 2025

2026

EXPERIENCE

MQI
Course Instructor
• Improved student performance by mentoring 8 students in structured class environment
• Increased knowledge retention and engagement, by applying proven memorization methods across 178+ pages of material

Fresh Prints
Sales Associate
• Increased sales by optimizing pricing strategies and managing client requirements.
• Handled multiple client interactions, improving communication and problem-solving skills.

Sep 2024 – Apr 2026

Willowbrook, IL

Oct 2023 – Mar 2024

Chicago, IL